

Heah Hung Xun

B.Sc. Physics (Year 2), Xiamen University Malaysia — mathematical physics: geometry, spectral methods, quantum foundations.

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Research

Petrov–Weyl spectral bounds

Independent work — unpublished draft

- Applied Gershgorin circles and Brauer Cassini ovals to eigenvalue localisation for the structured Weyl matrix in the Newman–Penrose formalism; introduced a candidate radiation-asymmetry invariant σ .

Anisotropic kernel control on the oriented frame bundle of S^2

AI-assisted exploratory draft

- Developed five propositions, one theorem, three conjectures, and two algorithms connecting differential geometry, Lie groups, geometric control, and approximation on manifolds.
- Numerically verified the predicted holonomy at $\theta_0 = \pi/4$; conjectural claims remain unproved and the manuscript is not submitted.

Gaussian quantum dynamics on $\text{Sp}(4, \mathbb{R})$

Ongoing undergraduate research

- Supervised by Prof. Tomasz Paterek. Use covariance-matrix and symplectic methods, Floquet theory for a periodically modulated Hamiltonian, logarithmic negativity, and periodic Riccati integration.
- Independently reproduced the numerical behaviour in Poddubny et al., Fig. 2, as a validation benchmark.

Quantum mechanics over non-complex number systems

Self-study note

- Compare real and quaternionic division-algebra formulations with completions of $\mathbb{Q}$, including p -adic models, to isolate which algebraic and topological assumptions standard quantum theory uses.

Lorentz–Abraham–Dirac to Landau–Lifshitz deviation

Course term paper

- Treated the reduction as a singular perturbation: τ_0 multiplies the highest derivative, so the $\tau_0 \rightarrow 0$ limit changes the order of the electrodynamic equation; studied the resulting deviation numerically.

Education

B.Sc. in Physics, Xiamen University Malaysia

2023–present

GPA: 3.75/4.00

Selected coursework: Quantum Mechanics I, Electrodynamics, Mathematical Methods, Theoretical Mechanics, Linear Algebra, Differential Equations, Numerical Methods.

Mathematical Methods

Eigenvalue localisation (Gershgorin, Cassini) · symplectic and covariance-matrix methods · Floquet theory · Hamilton–Jacobi and Madelung formulations · differential geometry, principal and frame bundles, holonomy · Lie groups · spherical harmonics · singular perturbation theory · division algebras and p -adic analysis.

Computational

Languages and tools: Python (NumPy, SciPy, QuTiP), Octave/MATLAB, C++, L^AT_EX/TikZ. **Numerics:** stiff integrators (Radau, BDF), Riccati integration, validation-first numerical workflows.

Other: Published poet.